

A corrected note on the TSP constant in Euclidean traveling-repairman latency

Prepared as a verification and repair of an attempted proof

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Abstract

Blanchard, Jacquillat and Jaillet asked whether the exact leading universal constant in the planar i.i.d. Euclidean traveling-repairman problem is the Beardwood–Halton–Hammersley constant β_{TSP} . The attempted proof that motivated this note is not correct: its main “BHH theorem for capped linear thinnings” uses a false lower-semicontinuity principle, and its stability argument is not justified as stated. This note records what can be salvaged rigorously. We prove the sharp β_{TSP} upper bound for the TRP for arbitrary compactly supported sampling laws, including singular parts. We also prove the corresponding upper bound for linear-cardinality partial TSP. Finally, we isolate a precise capped partial-TSP lower-bound conjecture and prove that this conjecture would imply the desired exact TRP asymptotic. Thus the present note gives partial progress and a clean conditional reduction, but it does not solve the open problem.

1 Introduction and verdict

Let $X_1, \dots, X_n$ be i.i.d. points in a compact set $K \subset \mathbb{R}^2$. The traveling-repairman objective, with a free starting point, is

$$L_n := l_{\text{TRP}}(X_1, \dots, X_n) = \min_{(x_1, \dots, x_n)} \sum_{i=1}^n \sum_{j < i} \|x_{j+1} - x_j\|.$$

Blanchard, Jacquillat and Jaillet prove constant-factor estimates for L_n , showing in particular that L_n is of order $n\sqrt{n}$ in expectation and with high probability [2]. In their conclusion they note that replacing their local space-filling construction by exact TSP subroutines improves the upper-bound constant from 2 to β_{TSP} , and they ask whether this β_{TSP} is tight [2, p. 1188]. This is the problem considered here.

The answer obtained in this note is:

There are serious mathematical errors in the attempted proof. I do not obtain a solution of the open problem. I obtain partial progress: a rigorous sharp upper bound with constant β_{TSP} , a sharp upper bound for linear partial TSP, and a conditional theorem reducing the full TRP constant question to a capped linear partial-TSP lower bound.

The obstruction is not cosmetic. The attempted proof assumes, in effect, that an adversarially chosen subset of qn points from an i.i.d. cloud satisfies the same BHH lower bound as an i.i.d. sample from its weak empirical limit. Weak convergence alone is insufficient; microscopic clustering can lower TSP length without changing the weak limit. See [Section 6](#) below.

2 Notation and the conjectural constant

Let the sampling law be

$$\mu = f \mathcal{L}^2 + \mu_s, \quad m := \int_K f(x) \, dx, \quad s := \mu_s(K) = 1 - m.$$

The singular part is allowed. The convention, consistent with the BHH theorem, is that the singular part contributes zero two-dimensional TSP length at the $\sqrt{n}$ scale.

For $a, b \geq 0$ define

$$H(a, b) := \mathbf{1}_{\{b < a\}} + \frac{1}{2} \mathbf{1}_{\{b = a\}}.$$

The distribution-dependent functional appearing in the TRP bounds is

$$\Gamma(f) := \iint_{K \times K} \sqrt{f(x) f(y)} H(f(x), f(y)) \, dx \, dy. \quad (1)$$

Since K is bounded and $f \in L^1(K)$, the integral is finite by Cauchy–Schwarz.

For the partial-TSP formulation, let

$$P_n(k) := \min\{\text{TSP}(A) : A \subset \{X_1, \dots, X_n\}, |A| = k\},$$

where $\text{TSP}(A)$ denotes the shortest Euclidean path through the finite multiset A . Path and cycle versions differ by at most $O(1)$ on a fixed compact set and therefore have the same $\sqrt{n}$ asymptotics.

For $0 \leq a \leq m$, define the capped-density functional

$$\psi_f(a) := \inf \left\{ \int_K \sqrt{\rho(x)} \, dx : 0 \leq \rho \leq f \text{ a.e., } \int_K \rho(x) \, dx = a \right\}. \quad (2)$$

Set

$$\Psi_f(q) := \begin{cases} 0, & 0 \leq q \leq s, \\ \psi_f(q - s), & s < q \leq 1. \end{cases} \quad (3)$$

The bathtub principle says that $\psi_f(a)$ is obtained by filling the highest-density parts of K first, with partial filling only on a threshold level of f .

Lemma 1 (Layer-cake identity). *One has*

$$\Gamma(f) = \int_0^1 \Psi_f(q) \, dq = \int_0^m \psi_f(a) \, da. \quad (4)$$

Proof. For a simple density

$$f = \sum_{j=1}^r \lambda_j \mathbf{1}_{A_j}, \quad \lambda_1 > \lambda_2 > \dots > \lambda_r > 0,$$

put $p_j = \lambda_j |A_j|$, $b_j = |A_j| \sqrt{\lambda_j}$ and $M_j = \sum_{\ell > j} p_\ell$. The bathtub minimizer fills $A_1, A_2, \dots$ in that order. While the level A_j is being filled, the value of ψ_f starts at $\sum_{\ell < j} b_\ell$ and has slope $1/\sqrt{\lambda_j}$. Integrating over the mass interval of length p_j assigned to A_j gives

$$\int_0^m \psi_f(a) \, da = \sum_{j=1}^r b_j \left(M_j + \frac{p_j}{2} \right).$$

The right side of (1) gives the same sum, because a point in A_j is paired with all lower-density mass and with one half of the equal-density mass.

For general f , truncate the set $\{f > R\}$ so that its contribution to the integrable kernel $(x, y) \mapsto \sqrt{f(x)f(y)}$ is arbitrarily small, then quantize $f \wedge R$ into intervals of mesh h . The preceding simple-density identity applies to the quantized functions. As $h \downarrow 0$ and then $R \uparrow \infty$, dominated convergence gives convergence of the double integrals in (1); the tie convention $1/2$ is exactly the symmetric limit on level sets. The variational formula (2) is stable under the same monotone quantization by the bathtub principle. This proves the identity. $\square$

3 A sharp upper bound for the TRP

We first state a deterministic space-filling estimate used repeatedly.

Lemma 2 (Space-filling bound). *For every compact $K \subset \mathbb{R}^2$ there is a constant C_K such that every multiset of $r \geq 1$ points in K can be visited by a Euclidean path of length at most $C_K\sqrt{r} + C_K$. Consequently its TRP latency is at most $C_K r\sqrt{r} + C_K r$.*

Proof. Enclose K in a square of side length D . Divide the square into $\lceil \sqrt{r} \rceil^2$ congruent subsquares and traverse the grid in a snake order, making a detour inside each occupied subsquare. The total length is at most a constant depending on D times $\sqrt{r} + 1$. The latency bound follows by multiplying path length by the number of visited points. $\square$

Theorem 1 (Sharp TRP upper bound). *For every compactly supported sampling law $\mu = f\mathcal{L}^2 + \mu_s$,*

$$\limsup_{n \rightarrow \infty} \frac{L_n}{n\sqrt{n}} \leq \beta_{\text{TSP}}\Gamma(f) \quad \text{almost surely.} \quad (5)$$

Moreover,

$$\limsup_{n \rightarrow \infty} \frac{\mathbb{E}L_n}{n\sqrt{n}} \leq \beta_{\text{TSP}}\Gamma(f). \quad (6)$$

Proof. We first prove a finite-priority version. Let $A_1, \dots, A_r$ be a measurable partition of the absolutely continuous support, ordered in the sequence in which we will serve them. Put

$$p_j := \int_{A_j} f, \quad b_j := \int_{A_j} \sqrt{f}, \quad M_j := \sum_{\ell > j} p_\ell.$$

Serve all observations generated from the singular component first. By the BHH theorem applied to the normalized singular law, their TSP length is $o(\sqrt{n})$ almost surely; hence their total contribution to all latencies is $o(n\sqrt{n})$. The single connecting edge to the absolutely continuous part contributes only $O(n)$.

For each j , let $N_{j,n}$ be the number of absolutely continuous observations falling in A_j , and let $T_{j,n}$ be the length of the optimal TSP cycle through these observations. Conditional on $N_{j,n}$, the points in A_j are i.i.d. with density $f\mathbf{1}_{A_j}/p_j$ when $p_j > 0$. The BHH theorem, together with $N_{j,n}/n \rightarrow p_j$, gives

$$\frac{T_{j,n}}{\sqrt{n}} \rightarrow \beta_{\text{TSP}} b_j, \quad \frac{N_{j,n}}{n} \rightarrow p_j \quad \text{a.s.} \quad (7)$$

For $p_j = 0$ the same conclusion is trivial.

Inside a fixed TSP cycle of length T through N points, some choice of starting vertex and direction has internal latency at most $NT/2$: averaging over the N starting vertices, each edge

receives average latency weight $(N - 1)/2$. Therefore, after choosing the best start in each block and connecting the finitely many blocks by arbitrary edges, the constructed repairman route has total latency at most

$$\sum_{j=1}^r \left(\sum_{\ell > j} N_{\ell,n} + \frac{N_{j,n}}{2} \right) T_{j,n} + o(n\sqrt{n}), \quad (8)$$

where the $o(n\sqrt{n})$ includes the singular preliminary route and the finitely many inter-block edges. Dividing by $n\sqrt{n}$ and using (7) yields

$$\limsup_{n \rightarrow \infty} \frac{L_n}{n\sqrt{n}} \leq \beta_{\text{TSP}} \sum_{j=1}^r b_j \left(M_j + \frac{p_j}{2} \right) \quad \text{a.s.} \quad (9)$$

It remains to choose the partition. Quantize the values of f into finitely many levels and order the cells by decreasing quantized density. As the quantization mesh tends to zero, after truncating large values of f if necessary, the sum on the right side of (9) converges to $\Gamma(f)$ by the same dominated-convergence argument as in Lemma 1. Thus, for every $\varepsilon > 0$, a finite ordered partition can be chosen so that the right side of (9) is at most $\beta_{\text{TSP}}\Gamma(f) + \varepsilon$. Letting $\varepsilon \downarrow 0$ gives (5).

The expectation bound follows from the same construction using the L^1 version of the BHH theorem and the law of large numbers for the counts. Indeed, for a fixed finite partition, take expectations in (8); uniform integrability is supplied by Lemma 2, and the local terms converge in L^1 to the right side of (9). Then use the same density-level partition approximation and let the partition error tend to zero. $\square$

Corollary 1 (Uniform law). *If μ is uniform on a compact region of area A , then*

$$\limsup_{n \rightarrow \infty} \frac{\mathbb{E}L_n}{n\sqrt{n}} \leq \frac{\beta_{\text{TSP}}}{2} \sqrt{A}.$$

The conjectured exact constant in this case is therefore $\beta_{\text{TSP}}\sqrt{A}/2$.

4 A sharp upper bound for linear partial TSP

The same ingredients give the easy half of the capped partial-TSP asymptotic.

Theorem 2 (Capped partial-TSP upper bound). *For every $q \in [0, 1]$,*

$$\limsup_{n \rightarrow \infty} \frac{P_n(\lfloor qn \rfloor)}{\sqrt{n}} \leq \beta_{\text{TSP}} \Psi_f(q) \quad \text{almost surely and in } L^1. \quad (10)$$

Proof. If $q \leq s$, select the required points from the singular observations whenever possible. The TSP length of all singular observations is $o(\sqrt{n})$ by BHH with zero absolutely continuous density. At $q = s$, if the singular count is short by $o(n)$, add $o(n)$ arbitrary points; by Lemma 2 this costs $o(\sqrt{n})$. Hence the limsup is zero for $q \leq s$.

Now assume $q > s$ and put $a = q - s$. Fix $\varepsilon > 0$ and choose a measurable density ρ with $0 \leq \rho \leq f$, $\int \rho = a$, and

$$\int_K \sqrt{\rho} \leq \psi_f(a) + \varepsilon.$$

Retain each original observation at location x from the absolutely continuous part independently with probability $\rho(x)/f(x)$, with the convention that this ratio is zero when $f(x) = 0$. The retained

observations have unconditional intensity density ρ ; their count is $an + o(n)$ almost surely, and conditional on being retained they are i.i.d. with density ρ/a . BHH gives a path through the retained observations of length

$$\beta_{\text{TSP}} \sqrt{n} \int_K \sqrt{\rho} + o(\sqrt{n}) \quad \text{a.s. and in } L^1.$$

Add the singular observations, whose total TSP length is $o(\sqrt{n})$, and add or delete $o(n)$ points to obtain exactly $\lfloor qn \rfloor$ points. Adding $o(n)$ points costs $o(\sqrt{n})$ by [Lemma 2](#); deleting points cannot increase the optimal path length. Thus

$$\limsup_{n \rightarrow \infty} \frac{P_n(\lfloor qn \rfloor)}{\sqrt{n}} \leq \beta_{\text{TSP}}(\psi_f(a) + \varepsilon).$$

Let $\varepsilon \downarrow 0$. □

5 A conditional solution of the TRP constant question

The lower bound for TRP is exactly where the attempted proof fails. The following deterministic observation is correct and useful.

Lemma 3 (Prefix reduction). *For every realization,*

$$L_n \geq \sum_{k=1}^n P_n(k). \quad (11)$$

Proof. In any repairman order $x_1, \dots, x_n$, the prefix $x_1, \dots, x_k$ is a path through k sampled points. Its length is therefore at least $P_n(k)$. Summing the n prefix lengths gives the TRP objective of that order. Taking the minimum over orders preserves the inequality. □

Assumption 1 (Integrated capped partial-TSP lower bound). *For the i.i.d. sample from μ ,*

$$\liminf_{n \rightarrow \infty} \frac{1}{n\sqrt{n}} \sum_{k=1}^n P_n(k) \geq \beta_{\text{TSP}} \int_0^1 \Psi_f(q) dq = \beta_{\text{TSP}} \Gamma(f) \quad \text{a.s.} \quad (12)$$

A stronger but more transparent sufficient condition is the pointwise lower bound

$$\liminf_{n \rightarrow \infty} \frac{P_n(\lfloor qn \rfloor)}{\sqrt{n}} \geq \beta_{\text{TSP}} \Psi_f(q) \quad \text{for Lebesgue-a.e. } q \in [0, 1]. \quad (13)$$

Indeed, (13) implies (12) by Fatou's lemma, after viewing the Riemann sum as an integral over q .

Theorem 3 (Conditional exact TRP constant). *If [Assumption 1](#) holds for μ , then*

$$\frac{L_n}{n\sqrt{n}} \longrightarrow \beta_{\text{TSP}} \Gamma(f) \quad \text{almost surely.} \quad (14)$$

The convergence also holds in L^1 , and hence

$$\mathbb{E} L_n \sim \beta_{\text{TSP}} \Gamma(f) n\sqrt{n}. \quad (15)$$

In particular, a proof of the capped partial-TSP lower estimate (13) would solve the Blanchard–Jacquillat–Jaillet constant question affirmatively.

Proof. The lower bound follows from [Lemma 3](#) and [Assumption 1](#). The upper bound is [Theorem 1](#). This proves almost-sure convergence. Uniform boundedness of $L_n/(n\sqrt{n})$ from [Lemma 2](#) then gives L^1 convergence by dominated convergence, and the expectation asymptotic follows. □

6 Why the original proof cannot be repaired locally

The attempted proof invoked a lower-semicontinuity principle of the form

$$\nu_n := \frac{1}{n} \sum_{x \in A_n} \delta_x \Rightarrow \rho(x) dx \implies \liminf_{n \rightarrow \infty} \frac{\text{TSP}(A_n)}{\sqrt{n}} \geq \beta_{\text{TSP}} \int \sqrt{\rho}. \quad (16)$$

This implication is false without additional microscopic regularity hypotheses.

Proposition 1 (Weak convergence alone does not control TSP length). *There are deterministic multisets $A_n \subset [0, 1]^2$ such that*

$$\frac{1}{n} \sum_{x \in A_n} \delta_x \Rightarrow \mathcal{L}^2|_{[0,1]^2}, \quad \limsup_{n \rightarrow \infty} \frac{\text{TSP}(A_n)}{\sqrt{n}} \leq \frac{\beta_{\text{TSP}}}{\sqrt{2}} < \beta_{\text{TSP}}.$$

Thus (16) is false.

Proof. Choose a deterministic sequence $z_1, z_2, \dots$ in $[0, 1]^2$ for which the empirical measures of $z_1, \dots, z_m$ converge to Lebesgue measure and

$$\frac{\text{TSP}(z_1, \dots, z_m)}{\sqrt{m}} \rightarrow \beta_{\text{TSP}};$$

almost every i.i.d. uniform sequence has these properties by the BHH theorem. For $n = 2m$, one may simply let A_{2m} contain two copies of each z_i , $1 \leq i \leq m$. If one insists on honest sets rather than multisets, choose a point $c \in (0, 1)^2$ not equal to any z_i and put

$$w_i^{(m)} = (1 - m^{-3})z_i + m^{-3}c, \quad A_{2m} = \{z_i, w_i^{(m)} : 1 \leq i \leq m\}.$$

The companion points stay in the square, are distinct from their bases, and satisfy $\max_i \|w_i^{(m)} - z_i\| = O(m^{-3})$. The empirical measures still converge to Lebesgue measure. A path can visit each pair consecutively, so

$$\text{TSP}(A_{2m}) \leq \text{TSP}(z_1, \dots, z_m) + O(m \cdot m^{-3}) = \text{TSP}(z_1, \dots, z_m) + o(\sqrt{m}).$$

Dividing by $\sqrt{2m}$ gives the claimed upper bound. Odd n are handled by deleting one point. $\square$

Random i.i.d. samples do not contain such exact pairs at macroscopic frequency, but [Proposition 1](#) shows that the missing input is precisely a no-clustering, sample-dependent subset lower bound. Proving such a bound for optimizing subsets of an i.i.d. cloud is the substantive capped partial-TSP problem, not a routine corollary of BHH.

The attempted proof also used a stability estimate with an unjustified assertion that an optimal repairman route has final length $O(\sqrt{n})$. The following weaker deterministic estimate is valid.

Lemma 4 (A valid but weaker stability estimate). *There is $C_K < \infty$ such that if two n -point multisets $A, B \subset K$ differ in at most r points, then*

$$|l_{\text{TRP}}(A) - l_{\text{TRP}}(B)| \leq C_K n \sqrt{r} + C_K n. \quad (17)$$

Consequently, if $r \leq \delta n$, then the scaled difference is $O_K(\sqrt{\delta})$.

Proof. Let $I = B \setminus A$ be the inserted multiset and $C = A \cap B$ the common multiset, with $|I| \leq r$. Start with a space-filling path through I , of length $O_K(\sqrt{r} + 1)$, connect to the first point of an optimal route for A with the deleted points skipped, and then follow that skipped route through C . The inserted points have total latency $O_K(r\sqrt{r} + r)$. Every common point is delayed by at most $O_K(\sqrt{r} + 1)$ compared with the skipped route, for a total additional latency $O_K(n\sqrt{r} + n)$. This gives $l_{\text{TRP}}(B) \leq l_{\text{TRP}}(A) + C_K n\sqrt{r} + C_K n$. Interchanging A and B proves the reverse inequality. $\square$

This corrected stability lemma is enough for some approximation arguments because $\sqrt{\delta} \rightarrow 0$, but it does not repair the missing capped partial-TSP lower bound.

7 Conclusion

The sharp β_{TSP} upper bound for TRP is valid and follows from density-ordered local TSP tours. The prefix inequality reduces the matching lower bound to an integrated linear partial-TSP lower bound. The attempted proof of that partial-TSP lower bound is invalid, and a direct weak-convergence argument cannot work because of microscopic clustering. Therefore the current status of this note is partial progress, not a resolution of the Blanchard–Jacquillat–Jaillet open problem.

References

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